

Finally, it's tested in a simulation to see if it really does what it's supposed to.

And now, it's off to the fabrication plant...

In The Clean Room

Even the smallest speck of dust can turn an IC into a worthless piece of silicon, so everything we're talking about here takes place in a "cleanroom," which is, well, a *very* clean room. We're talking millions of dollars worth of dust filters, special furniture that won't leave particles, even specialised cleaning materials.

It all starts with a wafer of pure silicon; the wafers are rinsed in pure water and a special cleaning solution. The huge number of transistors that will be made on this chip are going to be packed together real close, so it's necessary to isolate them from each other using an insulating material. Rather than slap on a layer of material on the silicon wafer, the wafer is exposed to heat and oxygen to create a layer of silicon dioxide on it. Now, we'll remove the oxide only from those areas that we need to create the transistors on.

The layer of silicon dioxide is then coated with a material called *photoresist*, which turns into an incoherent goo when exposed to ultraviolet light. Here's where you use the circuit diagram that came out of the design process. The circuit's layout is etched on to a *mask*, which is then placed on the layer of photoresist. This is then exposed to ultraviolet light, which causes the photoresist under the mask's transparent areas (which is where the transistors need to be) to turn gooey so it can be washed away. The wafer is then doused in acid, which eats up the exposed portions of the silicon dioxide layer, giving us access to the silicon underneath; this is called *etching*.

Now, through a process called *ion implantation*, ions are diffused into the silicon to create transistors. Note that we're talking *only* transistors here—even resistors and capacitors are created using transistors connected in special configurations; the gory details are too much to fit in here.

So we've created our components on the wafer; now it's time to bring in the wiring.

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Connecting Them All

We use the term "wiring" rather loosely, but we *are* following the intention. The components have been created, but they're not connected. The first thing to do is to close up the transistors inside insulating material—more silicon dioxide. This time, the process of masking and etching is repeated, only the mask is creating windows that lead to the transistors' terminals. A layer of metal is deposited now, which establishes the connections.

This is only the first layer of metal, which establishes only part of the connections that need to be made—another cycle of insulating, masking, etching and metal depositing follows, creating the next set of connections—this goes on for up to twenty layers. This is how they're able to connect so many components in such a small area.

And speaking of small...

The Nanometre Kerfuffle

If you've been following technology news for the past few months—specifically the antics of the processor giants—you've most likely encountered talk about the old 90 nm (nanometre) process, the on-its-way-out 65 nm and the new wave of 45 nm processes. What are these numbers and what's the big deal?

To put it in non-engineering terms, these numbers represent the effective length of the transistors on the chip—so in the same chip area, it's possible to put more 45 nm transistors than 60 nm. However, it's not just that bit that's causing the hype—it's the transistors themselves (more specifically, the transistors that you'll see on Intel's upcoming Penryn and Nehalem processors). The transistors are called *high-k metal gate transistors*, and their design reduces the amount of current wasted in the transistor, which in turn pulls down the power they consume and the amount of heat dissipated by them. A more detailed explanation would fill up a few issues of this magazine, so we won't go that deep.

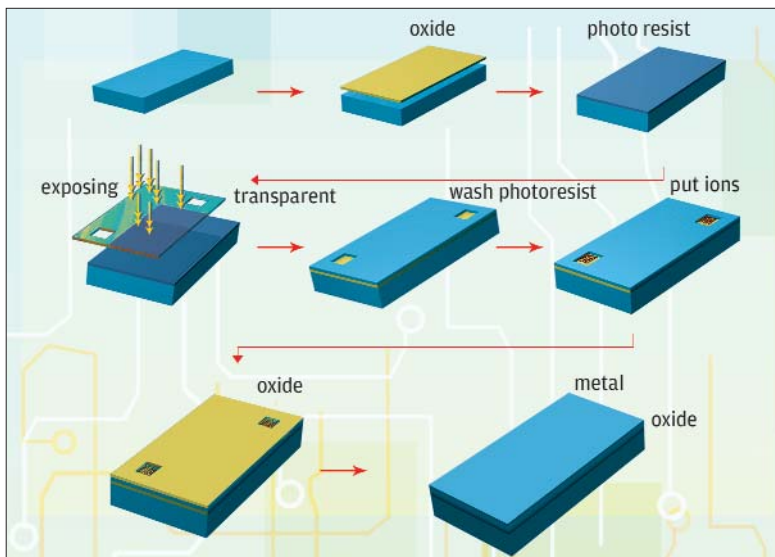
Ahead

Even with the new technologies, processor designs are getting more complex—HDLs have become cumbersome, and designers now prepare their specifications in variants of the C programming language. Inevitably, though, even C is going to fall short in the face of the complexity of newer hardware, so manufacturers are looking to newer methods to make processor design easier.

One way is to modularise the processor—break it into parts in such a way that they can be added or removed as necessary. An approach that looks promising is AMD's Fusion, which enables them to give you processors with as many CPU or graphics cores you want on the same chip. With the earlier method, if they wanted graphics and CPU on the same chip, they'd have to design it from the ground up—with Fusion, all they need to do is add the extra graphics core to an existing CPU core design, and *voilà*!

And then there's talk of offloading even *more* of the design to computers. Who knows, maybe we'll soon see processors whose sole purpose is to design better processors... ☒

nimish_chandiramani@thinkdigit.com



The process we've described above—how we start with a piece of silicon, go through the process, and end up with circuits etched on top of silicon dioxide